SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I K. Sasidhar Reddy (192525140) (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Sasidhar Reddy

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Define cloud computing and explain and four characteristics that make it different from traditional on premise computing

cloud computing is the delivery of computing resources such as servers, storage, database, software and networking over the Internet

Four characteristics

i) on-demand self-service

users can create or use computing resources whenever needed without direct assistance from the service provider

ii) Broad network access

cloud services can be accesed through the Internet using devices such as laptops, smartphones and tablets

iii) Resource Pooling

cloud provides share computing resources among multiple users using virtualization while maintaing proporer security and isolation

(iv) Rapid elasticity

Resources can be quickly increased or decreased according to demand. For example an online shopping website can add servers during a festival sales

(2) Trace the evolution of cloud computing from hardware evolution to Internet software evolution and show how these changes enabled modern cloud platforms

Cloud computing developed through several important stages:

Mainframe computing → client server → computing → virtualization → Internet/web services → cloud computing

(i) Mainframe computing

In the early stage, many users shared a powerful central mainframe computer through terminals

(ii) client-server computing

computing responsibilities were divided b/w clients and servers. This allowed organized to provide applications and data through networked computers

(iii) Virtualization:

Virtualization allowed one physical server to run multiple virtual machines. This improved hardware utilization and reduced infrastructure costs

(iv) Modern Cloud Platforms

Cloud provides combined virtualization high-speed networks, distributed data centers, automation and web services to provide scalable services such as IaaS, PaaS, and SaaS

Explain the role of server virtualization in cloud computing Differentiate b/w virtualization as an enabling technology and cloud computing as a service model

Role of Server virtualization

Server virtualization divides one physical server into multiple virtual machines (VMs) using a software layer called a hypervisor

For example:

Physical server

↓

Hypervisor

↓

VM1 + VM2 + VM3

Benefits

* Improves physical server utilization
* Reduces hardware and electricity costs
* Makes resource allocation flexible
* Supports easy scaling and VM migration

virtualization	cloud computing
* Technology used to create virtual resources	* Service model for delivering computing resources
* Uses hypervision and virtual machines	* Uses data centers, networks virtualization and automation
* Ex: VMware virtual machine	* Ex: Amazon EC2

Discuss the role of data centers in delivering cloud services

A data center is a facility containing servers storage system, networking equipment, power used to provide computing services

they are five types

Roles of data centers

(i) computing resources

Data centers contain powerfull servers that run cloud applications and virtual machines

(ii) Data storage

They store user-file, databases, backups and application data

(iii) Networking

High-speed networks connect servers and provide Internet access to users

(iv) availability and reliability

Backup power, redundant hardware, and multiple network connections help keep services available

(v) Scalability

cloud providers can add more servers and storage when user demand increases

(3) Differentiable IaaS PaaS, SaaS and XaaS with one suitable real world example for each model

Model	meaning	wait it provides	Example
IaaS	Infrastructure as a service	virtual machines	Amazon EC2
PaaS	Platform as a service	Development Platforms	Google APP Engine
SaaS	software as a service	Read to use software through Internet	Google Docs
XaaS	Anything as a service	Different IT resources delivered as a service	Amazon RDS

Iaas:
The provider given basic infrastructure such as virtual servers, storage, and networking. The customer manages the operating system and applications

Paas:
The provider gives a ready-made platforms for developing and deploying applications. The developer mainly focuses on application development

Saas:
The complete software application is provided through the Internet users simply access and use the application

Xaas:
"Anything as a service" is a board concept where different IT capabilities are provided as services on demand

12/10/26

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks: